

Time Series Analysis on Unemployment Rate in Italy

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I. INTRODUCTION

The unemployment rate is a key measure of a country's economic well-being, offering insight into labor market conditions and the overall economic stability and trends. This project is focusing on the time series analysis of the monthly unemployment rate in Italy.

Our goal is to find the most suitable model for analyzing and forecasting this time series. To do so, we examine patterns in the data, including underlying trends and seasonal effects. We test a range of models, from classical statistical techniques to more flexible machine learning approaches, and compare their performance. Through this comparison, we aim to highlight the strengths and limitations of each method and identify the one that best captures the dynamics of Italy's unemployment rate.

II. DATASET SELECTION

The primary data set used in this study is the monthly total unemployment rate in Italy [1], which serves as the target variable for our time series analysis.

To provide a more comprehensive understanding of the dynamics influencing unemployment, we selected several complementary economic indicators:

- **Retail Trade Volume:** reflects consumer spending behavior and economic confidence. [2]
- **Harmonized Index of Consumer Prices (HICP):** represents inflation and purchasing power, which can influence employment levels. [3]
- **Production in Industry:** a proxy for overall economic activity and business cycles. [4]
- **EUR-USD Exchange Rate:** captures external economic conditions that may affect trade and industry employment. [5]

These variables were chosen based on their economic relevance and potential explanatory power for modeling and forecasting unemployment trends. All the datasets were originally taken from EUROSTAT website.

III. EXPLORATORY DATA ANALYSIS

This section presents a visual examination of the unemployment rate in Italy and its selected complementary

economic indicators to identify key characteristics, trends, and relationships. Figure 1 shows the unemployment rate in Italy over time, revealing clear cyclical behavior and potential seasonality, with recurring peaks that suggest the presence of trend and seasonal components. The ACF plot 2 displays slowly decaying positive autocorrelations, indicating strong temporal dependence and suggesting the series may be non-stationary. The ACF also shows significant spikes at lags 12, 24, 36, etc., indicating a strong seasonal component with a 12-month period.

The PACF plot 3 shows a sharp drop after the first lag, implying that an autoregressive model of order 1 (AR(1)) could be appropriate for capturing the underlying structure of the series. Together, these preliminary observations guide the choice of models that incorporate autoregressive and possibly seasonal components.

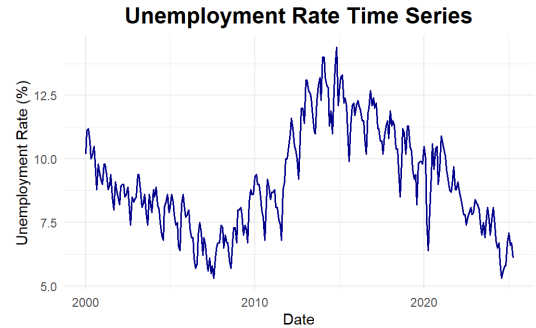


Figure 1: Unemployment Rate over Time

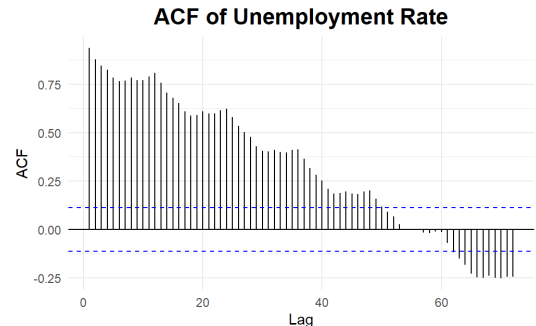


Figure 2: ACF for Unemployment Rate

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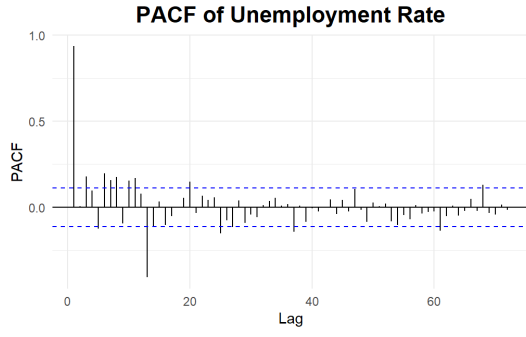


Figure 3: PACF for Unemployment Rate

IV. MODEL ANALYSIS

A. Linear Regression

We used a Linear Regression with quadratic trend and seasonality to forecast the unemployment rate.

$$Y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \sum_{j=1}^{11} \gamma_j S_{j,t} + \epsilon_t \quad (1)$$

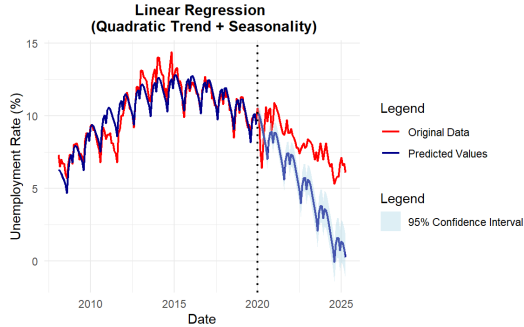


Figure 4: Linear Regression Forecast for Unemployment Rate

As we can see in Figure 4, we are filtering out data from before 2008 (global financial crisis), since this shock cannot be modeled with linear regression. We want to forecast the time series after 2020, which is when the effects of COVID-19 lockdown were beginning, and so it allows our model to forecast based on a period which is less influenced by external factors on a period more influenced by them.

The forecast, as we will see later in Table 1, has a very high RMSE and MAPE, compared to the training set, suggesting that the behavior of the data after COVID cannot be modeled with the same quadratic trend learned using data before COVID.

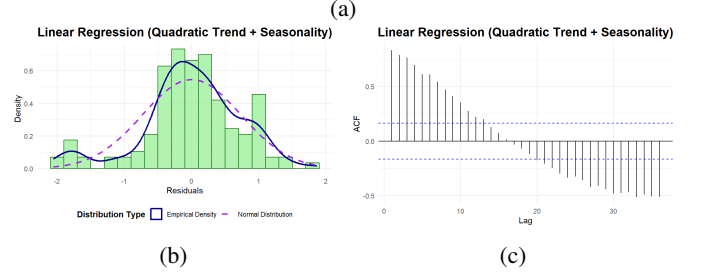
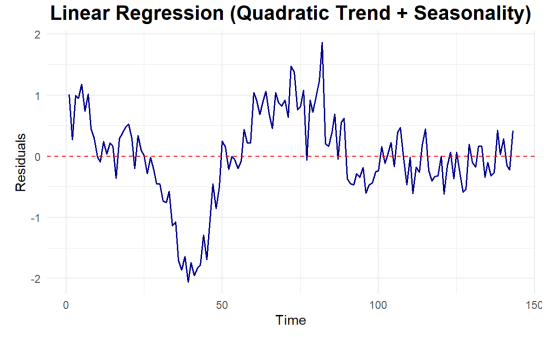


Figure 5: Linear Regression Model Residuals: (a) Time series plot of residuals, (b) Distribution of residuals, and (c) ACF of residuals.

On Figure 5c, we can see that the ACF for residuals of our quadratic linear regression for prediction on the training set is high at the start and gradually decreases to zero and then to negative values, which shows that the linear regression model has not captured the trend based on the ACF. We can also see the residuals themselves in 5a and how they are moving away from the baseline a bit too much, strongly approving that the model has not been able to model the trend. On the other hand we can see that the seasonality present in the original ACF from Figure 1 is hardly present now, pointing that the model has captured seasonality to some acceptable extent.

B. Holt-Winters Exponential Smoothing

Holt-Winters uses three smoothing equations, one for each component (level, trend, and seasonality), along with a forecast equation. Each equation uses a weighted average of past observations, with weights decaying exponentially as you go further back in time.

$$L_t = \alpha \left(\frac{Y_t}{S_{t-s}} \right) + (1 - \alpha)(L_{t-1} + b_{t-1}) \quad (2)$$

$$b_t = \beta(L_t - L_{t-1}) + (1 - \beta)b_{t-1} \quad (3)$$

$$S_t = \gamma \left(\frac{Y_t}{S_{t-s}} \right) + (1 - \gamma)S_{t-s} \quad (4)$$

$$F_{t+m} = (L_t + b_t m) S_{t-s+m} \quad (5)$$

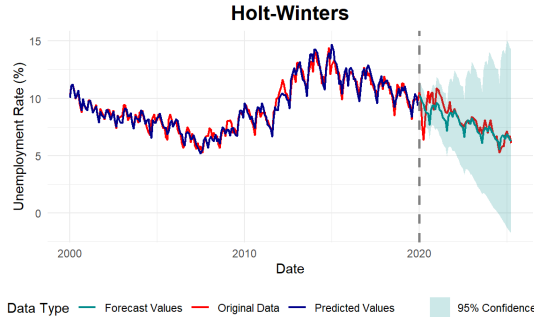


Figure 6: Holt-Winters Exponential Smoothing Forecast for Unemployment Rate

We are using multiplicative seasonality for our Holt-Winters model. We also tried additive seasonality but the performance was not as good as the current one. Even by looking at the time series, there is some correlation between seasonal fluctuations and trend and using multiplicative seasonality makes more sense.

The forecast is again focused on the data after 2020, like it was for linear regression and later will be for ARIMA.

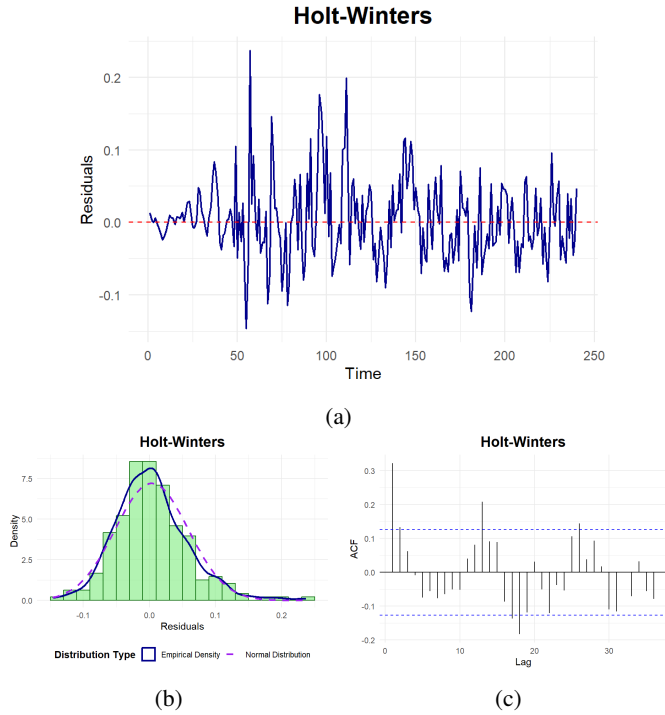


Figure 7: Holt-Winters Exponential Smoothing Model Residuals: (a) Time series plot of residuals, (b) Distribution of residuals, and (c) ACF of residuals.

The Holt-Winters model (Figure 5c) appears to follow the data more closely than the linear regression model. The residual analysis (Figure 7) shows a marked improvement. The residuals are more randomly scattered around zero, and the ACF plot (Figure 7c) shows fewer significant spikes, indicating better handling of the time series structure. Although,

there seems to be some seasonality to the residuals, but it is not following the yearly pattern of the seasonality of the original time series.

C. ARIMA

We chose ARIMA(1,1,0)(1,1,0)[12] by looking at our ACF (Figure 2) and PACF (Figure 3) of the original time series. The AR behavior is easily visible from these figures, since our PACF cuts off very fast and our ACF is gradually decreasing with a sinusoidal seasonality. Since the seasonality is strong, using ARIMA without seasonality does not lead to satisfying results, thus we also add the seasonality to our model.

- **(p,d,q) = (1,1,0)**: One non-seasonal autoregressive term ($p = 1$), one level of non-seasonal differencing ($d = 1$) to address the trend, and no non-seasonal moving average term ($q = 0$).
- **(P,D,Q)[12] = (1,1,0)[12]**: One seasonal autoregressive term ($P = 1$), one level of seasonal differencing ($D = 1$) for the 12-month seasonality, and no seasonal moving average term ($Q = 0$).

The model equation is:

$$(1 - \phi_1 B)(1 - \Phi_1 B^{12})(1 - B)(1 - B^{12})Y_t = \varepsilon_t \quad (6)$$

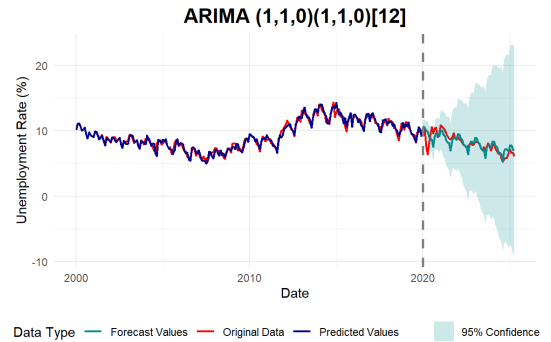


Figure 8: ARIMA Forecast for Unemployment Rate

As we can see in the Figure 9c, there are still a few significant autocorrelation values in our ACF plot, but compared to Holt-Winters, we have almost removed the seasonal pattern with our ARIMA model. Also it is easily seen from Figure 8 that the predicted values offer a very good fit to the data with an $RMSE = 0.4750$, and also a forecast with an $RMSE$ of 0.9367 .

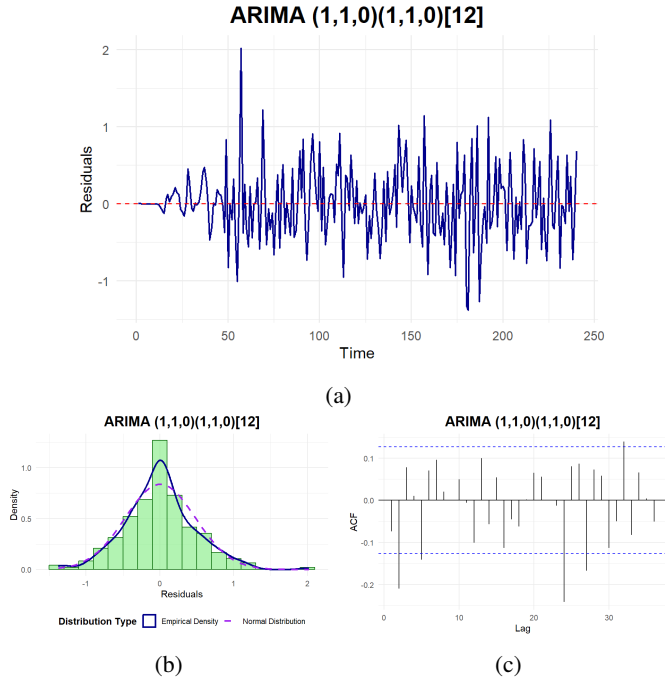


Figure 9: ARIMA Model Residuals: (a) Time series plot of residuals, (b) Distribution of residuals, and (c) ACF of residuals.

D. AUTO ARIMA

Our second ARIMA model, ARIMA(0, 1, 1)(0, 1, 1)[12], was chosen by auto.arima function from "forecast" library.

The model equation is:

$$(1 - B)(1 - B^{12})Y_t = (1 + \theta_1 B)(1 + \Theta_1 B^{12})\epsilon_t \quad (7)$$

The ARIMA(0, 1, 1)(0, 1, 1)[12] also provides a very good fit to the data, as shown in Figure 10. More importantly, its residuals (Figure 11a) appear to be white noise, with only two slightly significant autocorrelation values (Figure 11c), showing that the model has successfully captured the behavior of the data. Compared to our own ARIMA model, the fit to the training data is slightly better, with an $RMSE = 0.4421$, and the forecast is showing a nonsignificant, slightly poorer performance with an $RMSE = 0.9440$.

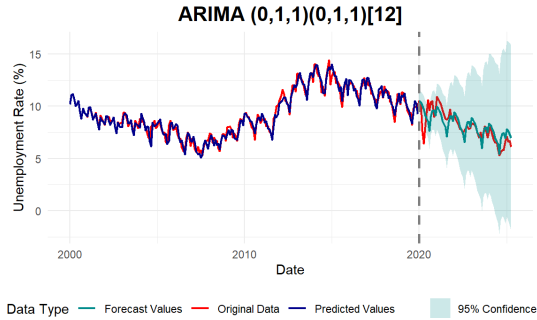


Figure 10: AUTO-ARIMA Forecast for Unemployment Rate

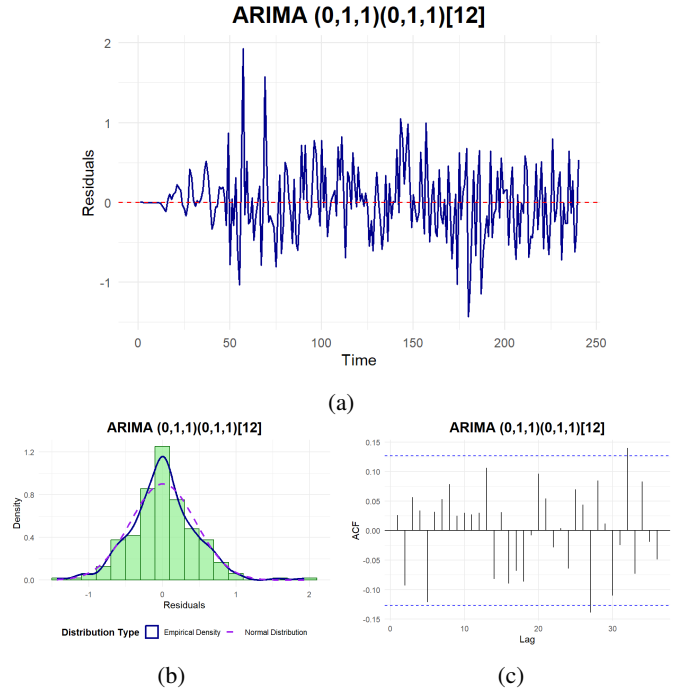


Figure 11: AUTO-ARIMA Model Residuals: (a) Time series plot of residuals, (b) Distribution of residuals, and (c) ACF of residuals.

E. Non-Parametric Regression

Non-Parametric Methods like LOESS, Smoothing Splines, and Generalized Additive Models (GAM) are used in this study to capture complex non-linear trends without strong parametric assumptions. While these models can be used for short-term forecasting under certain conditions, we use them here primarily for exploratory analysis, and not for forecasting. Their residuals (Figures 13, 17, 19) show significant autocorrelation, as these models do not capture characteristics such as lagged effects or seasonality. Therefore, despite their strengths in visualizing trends, their limitations in extrapolating over time with autocorrelated data make them less suitable than models like ARIMA and Holt-Winters for forecasting.

1) Local Regression

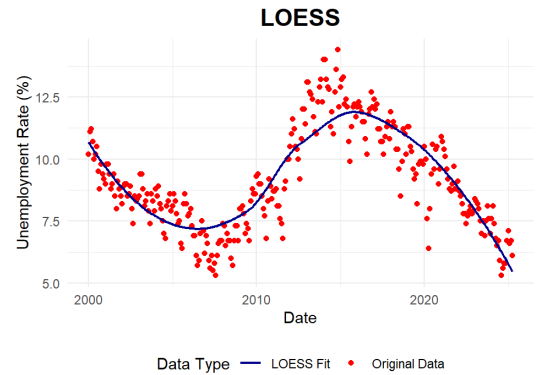


Figure 12: LOESS Prediction for Unemployment Rate

We use $Span = 0.75$ for our Local Regression Model, which is chosen by trial and error. As we can see, it is providing us with a very good overall trend but as mentioned before the residuals are not anything close to a white noise, since specially the seasonality is not addressed at all.

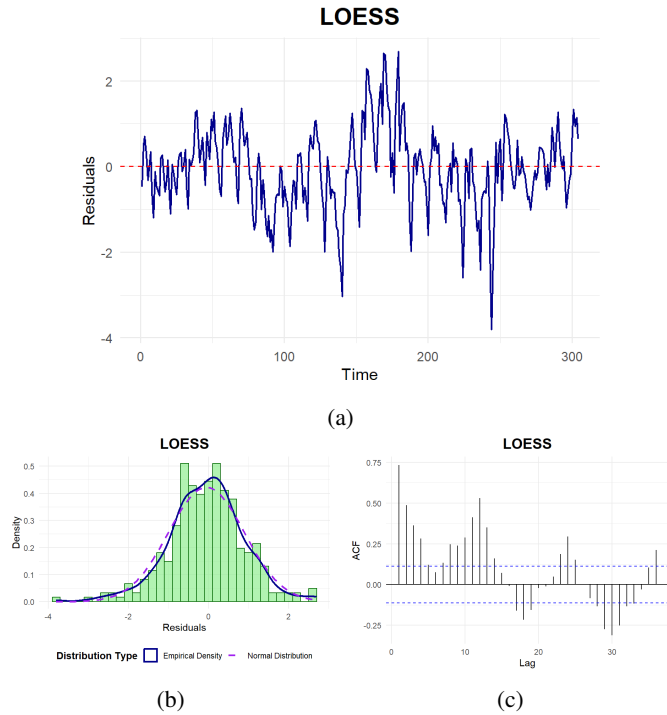


Figure 13: LOESS Model Residuals: (a) Time series plot of residuals, (b) Distribution of residuals, and (c) ACF of residuals.

2) Regression Splines

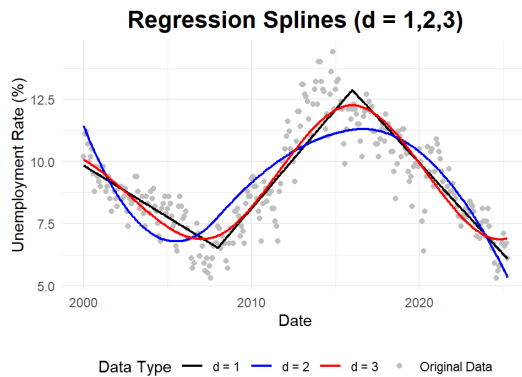


Figure 14: Regression Splines Prediction for Unemployment Rate ($d = 1, 2, 3$)

For Regression Splines, we use two knots and we place one on January 2008, for the global financial crisis and another one at 2016, to have placed them almost uniformly. Also as we can see in Figure 14, we have used $d = 1, 2, 3$ for the degree of our polynomials.

3) Smoothing Splines

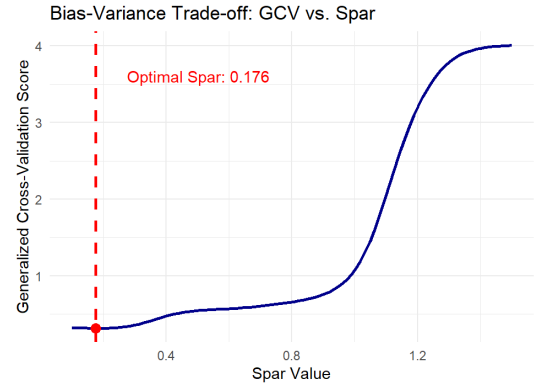


Figure 15: Finding the Optimum ($\lambda = 0.176$) Using Cross-Validation

To model with Smoothing Splines, we run a script to find the best value based on Bias-Variance Trade-off (Figure 15). The best λ as we can see in Figure 15, is 0.176, thus we proceed to use it for our model.

Of course we used other spar values to try to see how they act, but their performance was not worthy of mentioning here.

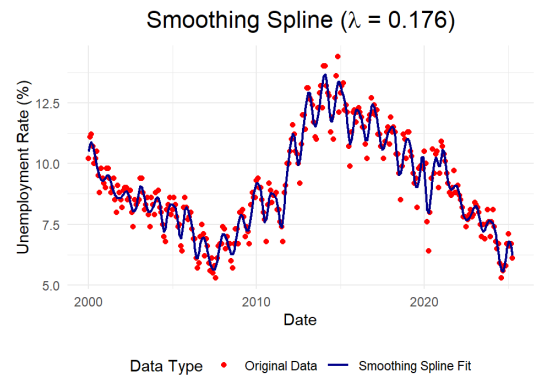


Figure 16: Smoothing Spline Prediction for Unemployment Rate ($\lambda = 0.176$)

Since the λ is small, we have some fluctuations in our predicted fit, but that is the cost to capture the trend much better than our other non-parametric models. It is mainly because other non-parametric models do not capture any seasonality, but here we are capturing some seasonality which is actually a yearly seasonality, thus minimizing the sinusoidal property of the residual autocorrelation.

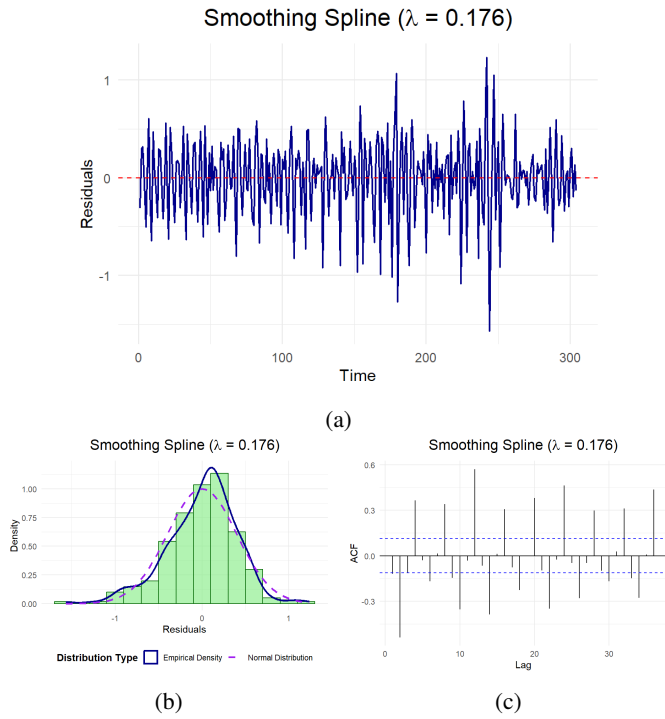


Figure 17: Smoothing Spline Model Residuals: (a) Time series plot of residuals, (b) Distribution of residuals, and (c) ACF of residuals.

4) GAM

Here we first use a very basic GAM model to see how it performs, then we make it more complex to see if it can capture the trend and seasonality of the data when we add other parameters to our additive model.

We can see from Figure 18 that the general fit is a good fit, but looking at the residuals in Figure 19 show that the model has not captured the seasonality, which was expected from a simple linear-like GAM model.

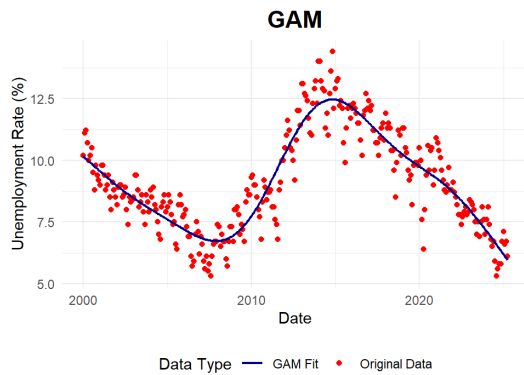


Figure 18: GAM Prediction for Unemployment Rate

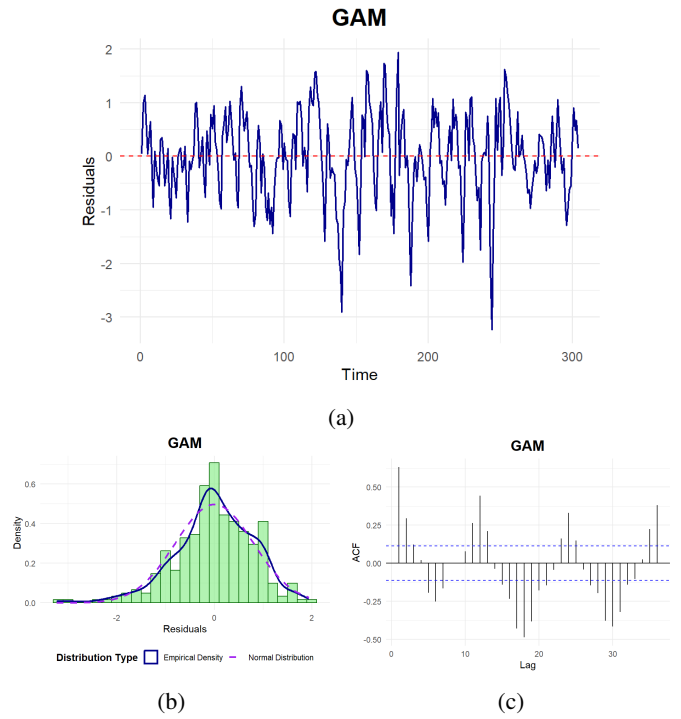


Figure 19: GAM Model Residuals: (a) Time series plot of residuals, (b) Distribution of residuals, and (c) ACF of residuals.

In addition to using the linear $UnemploymentRate = s(time\ index)$ GAM, we add our complementary data to our GAM to see how it improves the model. As we can see in Figure 20, the model now is much more complex, and also predicts the shock from COVID and 2008 financial crisis, since our complementary data is very much influenced by these two external factors.

As we look at the residuals from Figure 21, we can see that the spikes have a relatively similar pattern to our original GAM model, but the now the spikes are much less sharper than how they were before, indicating a better fit and a more white-noise like residuals.

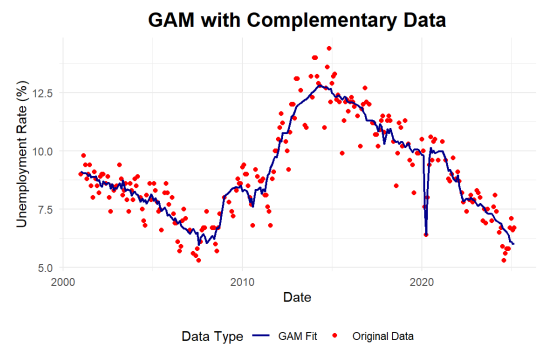


Figure 20: GAM with Complementary data Prediction for Unemployment Rate

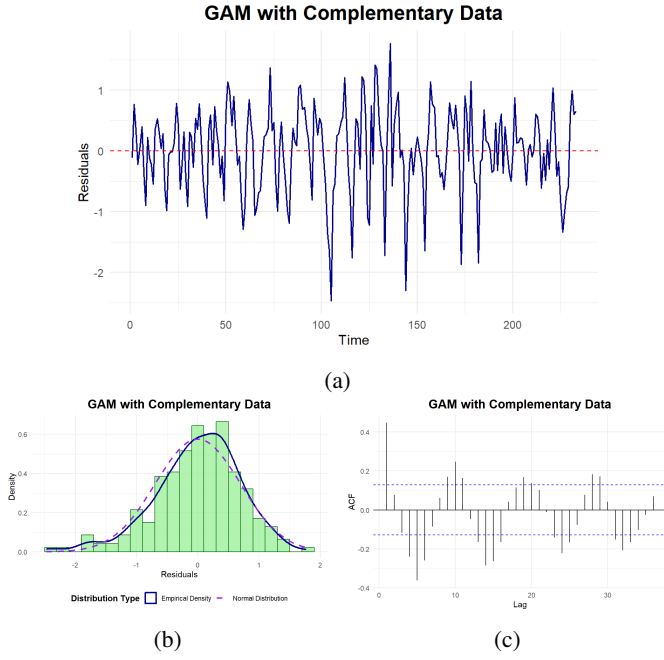


Figure 21: GAM Model with complementary data Residuals: (a) Time series plot of residuals, (b) Distribution of residuals, and (c) ACF of residuals.

F. Gradient Boosting

Gradient Boosting is a powerful algorithm that uses the external economic indicators as predictors. It builds a sequence of decision trees, where each tree corrects the errors of its predecessor. The model is built as:

$$F_m(x) = F_{m-1}(x) + \nu \sum_{j=1}^J \gamma_{jm} I(x \in R_{jm}) \quad (8)$$

Correlation Matrix (Response Variable: Unemployment Rate)

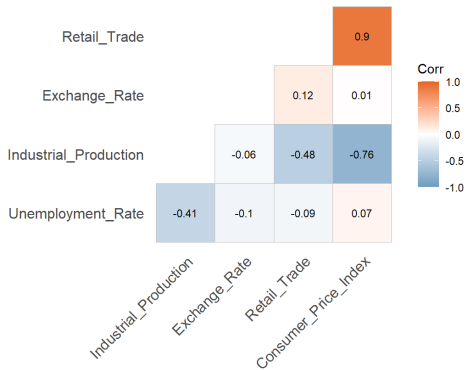


Figure 22: Correlation Matrix

We are using the complementary data introduced in Section II, and we can see the correlation matrix of these data and the original Unemployment Rate dataset, visualized in Figure 22. Even though these external parameters seem to not have strong linear influence on unemployment, we are going to use

a Gradient Boosting model to show how these data have an actual nonlinear correlation.

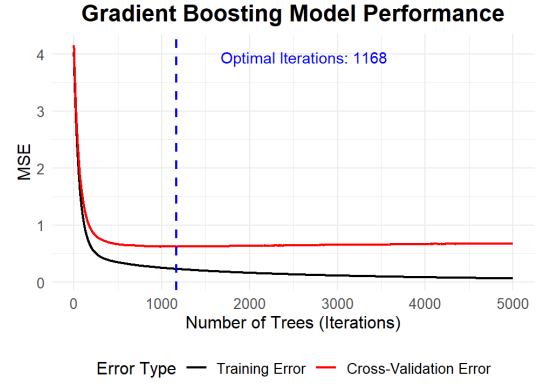


Figure 23

We use $number.of.trees(number.of.iterations) = 5000$ and $interaction.depth = 4$ with a $shrinkage = 0.01$ for our Gradient Boosting model, and we find the optimal Cross-Validation Error at $iteration = 1168$ as we can also see in Figure 23.

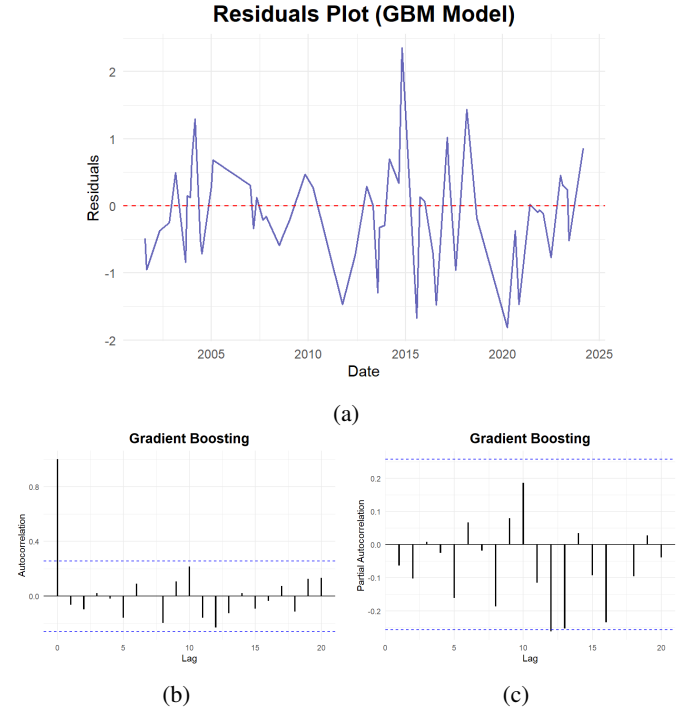


Figure 24: GB Model Residuals: (a) Time series plot of residuals, (b) ACF of residuals, and (c) PACF of residuals.

As we can see in Figure 24, the model is performing very well on our test set, virtually having no significant autocorrelation values or spikes in our ACF plot other than for the first order lag.

The variable importance plot (Figure 25) provides valuable insight to identify actual correlations, identifying the Con-

sumer Price Index as the most influential predictor by a large margin, followed by Retail Trade and Industrial Production.

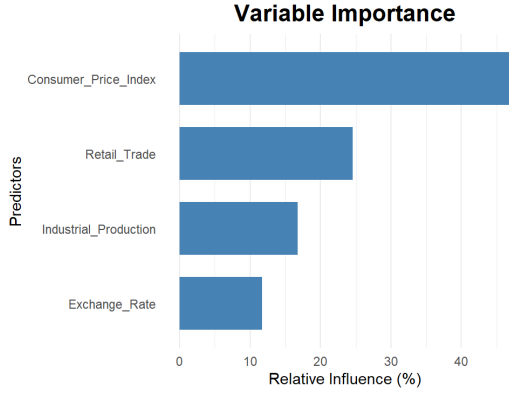


Figure 25: Variable Importance for GB

The partial dependence plots (Figure 26) further show non-linear relationships between these predictors and the unemployment rate. For example, the effect of the Consumer Price Index on unemployment appears to be non-monotonic, rising sharply before declining, suggesting that both very low and very high inflation levels may be associated with lower unemployment, while moderate inflation might correspond to higher unemployment.

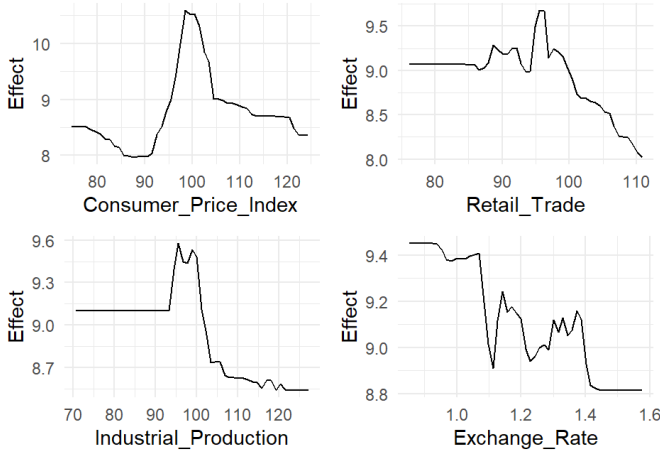


Figure 26: Partial Dependence Plots for GB

V. RESULTS

Performance of the primary forecasting models was quantitatively compared using standard metrics on both the training and testing datasets. The results are summarized in Table 1.

Linear Regression model is clearly inadequate, as evidenced by its extremely high RMSE (3.385) and MAPE (41.19) on the test set, indicating severe overfitting and an inability to generalize.

The auto ARIMA model demonstrates the best in-sample fit, achieving the lowest RMSE (0.4421) and MAPE (3.60) on the training data. This is further supported by its having the

Models	Data	RMSE	MAPE	AIC	BIC
Linear Regression	train	0.7271	5.67	344.69	389.13
	test	3.385	41.19	-	-
Holt-Winters	train	0.4873	4.11	1023.01	1082.18
	test	0.6560	7.66	-	-
AUTO-ARIMA	train	0.4421	3.60	297.28	307.56
	test	0.9440	11.35	-	-
ARIMA	train	0.4750	3.87	326.66	336.94
	test	0.9367	8.86	-	-
Gradient Boosting	train	0.4788	4.29	-	-
	test	0.7843	6.65	-	-

Table 1: Model Performance Comparison

lowest information criteria values (AIC: 297.28, BIC: 307.56), which penalize model complexity and reward goodness of fit. Also our ARIMA comes close as second, which is expected. However, the forecasting performance of both ARIMA models on the test set is weaker than that of Holt-Winters and Gradient Boosting.

The Gradient Boosting model achieves the best forecasting performance on unseen data, with the lowest MAPE (6.65) and a competitive RMSE (0.7843) on the test set. This highlights the significant advantage of incorporating relevant external predictors. The model was able to learn the complex interplay between the broader economy and unemployment, leading to more accurate predictions.

VI. CONCLUSION

This project compared several time series models to forecast Italy's monthly unemployment rate. While the Auto-ARIMA model provided the best fit on the training data, achieving the lowest RMSE of 0.4421 and the lowest information criteria (AIC: 297.28, BIC: 307.56), its forecasting performance was weaker on test data.

The Gradient Boosting model delivered the best forecasting performance on the test set, with the lowest MAPE of 6.65 and a strong RMSE of 0.7843. This highlights the significant advantage of incorporating external predictors, such as the Consumer Price Index, which was identified as the most influential variable. The results suggest that for complex economic data, a machine learning approach that uses additional relevant information can outperform traditional models like ARIMA and Holt-Winters in predicting future trends.

Future work can focus on hybrid models, or adding more comprehensive analysis of the effects of the parameters on each model, finding an even better solution to fit and forecast.

REFERENCES

- [1] EUROSTAT, "Euro area unemployment rate." https://ec.europa.eu/eurostat/databrowser/view/ei_lmhr_m/default/table?lang=en.
- [2] EUROSTAT, "Volume of retail trade." https://ec.europa.eu/eurostat/databrowser/view/sts_trtu_m/default/table?lang=en.
- [3] EUROSTAT, "Harmonised Indices of Consumer Prices (HICP)." https://ec.europa.eu/eurostat/databrowser/product/view/prc_hicp_midx?lang=en.
- [4] EUROSTAT, "Production in industry, howpublished = https://ec.europa.eu/eurostat/databrowser/view/sts_inpr_m/default/table?lang=en."
- [5] European Central Bank, "US dollar-euro reference exchange rate." <https://data.ecb.europa.eu/data/datasets/EXR/EXR.M.USD.EUR.SP00.E>.

VII. APPENDIX

We added the R summaries of models to the Appendix for more clarification.

```
Call:
lm(formula = Unemployment_Rate ~ time_index_temp + I(time_index_temp^2) +
    Month, data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.05738 -0.34631 -0.00367  0.44456  1.86183

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.066e+00  3.010e-01  20.150  < 2e-16 ***
time_index_temp  1.512e-01  6.253e-03  24.174  < 2e-16 ***
I(time_index_temp^2) -8.544e-04  4.206e-05 -20.312  < 2e-16 ***
MonthFeb      7.364e-02  3.199e-01   0.230  0.81831
MonthMar     -1.372e-01  3.199e-01  -0.429  0.66871
MonthApr     -5.047e-01  3.198e-01  -1.578  0.11705
MonthMay     -9.037e-01  3.198e-01  -2.826  0.00547 **
MonthJun     -1.276e+00  3.198e-01  -3.990  0.00011 ***
MonthJul     -1.680e+00  3.198e-01  -5.254  5.98e-07 ***
MonthAug     -2.399e+00  3.198e-01  -7.502  9.02e-12 ***
MonthSep     -9.746e-01  3.198e-01  -3.048  0.00280 **
MonthOct     -3.485e-01  3.198e-01  -1.090  0.27788
MonthNov     -1.790e-01  3.199e-01  -0.560  0.57669
MonthDec     -8.328e-01  3.199e-01  -2.603  0.01032 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7656 on 129 degrees of freedom
Multiple R-squared:  0.8703,    Adjusted R-squared:  0.8573
F-statistic: 66.61 on 13 and 129 DF,  p-value: < 2.2e-16
```

Figure 27: R summaries for Linear Regression

```
Holt-winters' multiplicative method

Call:
hw(y = train_ts, seasonal = "multiplicative")

Smoothing parameters:
alpha = 0.3067
beta = 0.019
gamma = 0.4775

Initial states:
l = 10.675
b = -0.0885
s = 0.9676 0.9887 1.0289 0.9147 0.9753 1.0547
      1.006 0.9701 1.0238 1.0698 1.0497 0.9508

sigma: 0.0572

      AIC      AICC      BIC
1023.007 1025.764 1082.178

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.0009611833 0.4873113 0.3768512 0.02867827 4.112752 0.4664608 0.3350279
```

Figure 28: R summaries for Holt-Winters

```
Series: train_ts
ARIMA(0,1,1)(0,1,1)[12]

Coefficients:
      mal      smal
-0.3997   -0.3768
s.e.    0.0643   0.0671

sigma^2 = 0.2085: log likelihood = -145.64
AIC=297.28  AICC=297.39  BIC=307.56

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.008305699 0.4420861 0.3254743 0.08888791 3.598432 0.4028672 0.02589265
```

Figure 29: R summaries for auto ARIMA

```
Call:
arima(x = train_ts, order = c(1, 1, 0), seasonal = c(1, 1, 0))

Coefficients:
      ar1      sar1
-0.3170   -0.3682
s.e.    0.0632   0.0616

sigma^2 estimated as 0.2385: log likelihood = -160.33, aic = 326.66

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.004506808 0.4749504 0.3513379 0.03437612 3.868384 0.6509283 -0.07388205
```

Figure 30: R summaries for ARIMA

```
Family: gaussian
Link function: identity

Formula:
Unemployment_Rate ~ s(time_index)

Parametric coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.13651    0.04666   195.8  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
              edf Ref.df      F p-value
s(time_index) 8.221  8.834 175.8  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.837   Deviance explained = 84.2%
GCV = 0.68253   Scale est. = 0.66183    n = 304
```

Figure 31: R summaries for GAM

```
Family: gaussian
Link function: identity

Formula:
Unemployment_Rate ~ s(time_index) + s(Industrial_Production) +
    s(Exchange_Rate) + s(Retail_Trade) + s(Consumer_Price_Index)

Parametric coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.09013    0.04693   193.7  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
              edf Ref.df      F p-value
s(time_index)  6.488  7.446 24.842  < 2e-16 ***
s(Industrial_Production) 4.362  5.413  4.836  0.000346 ***
s(Exchange_Rate)      1.000  1.000  0.772  0.380610
s(Retail_Trade)       6.904  7.903  1.358  0.195840
s(Consumer_Price_Index) 1.000  1.000  0.225  0.636104
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.875   Deviance explained = 88.6%
GCV = 0.56343   Scale est. = 0.51324    n = 233
```

Figure 32: R summaries for GAM + Complementary Data